

Nutrition Prediction based on Convolution Network

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Abstract—We address calorie estimation from paired color and depth images (raw and colorized), training all networks from scratch. The method uses a two-branch convolutional model: a depth branch to capture volume cues and a color branch to capture food composition. A lightweight fusion head with a learnable weight and an explicit interaction term combines both feature sets to predict calories. We follow an iterative design process with an integration of branch width and depth, pooling, and fusion strategies. The final model achieves a coefficient of determination of about 0.81 on the validation set. We provide visual diagnostics, including predicted versus target scatter plots and best and worst examples, plus a feature correlation study. Error analysis shows frequent underestimation for stacked or occluded dishes, motivating future work on depth preprocessing, occlusion aware augmentation, and improved fusion.

Index Terms—depth sensing, calorie estimation, multimodal fusion, convolutional neural networks, regression, error analysis

I. INTRODUCTION

Estimating physical quantities from images is a core aim of computer vision. In this research, we predict a meal's calories from color and depth images, emphasizing design reasoning over reuse of pre-trained models (all networks are randomly initialized). We separate geometry and appearance via depth and RGB branches, then fuse their features to produce a single regression output. This separation is motivated by the different roles of volume (depth) and composition (RGB). We document the step-by-step experiments that shaped the architecture and regularization choices, and report clear metrics and visualizations. Finally, we analyze errors, especially underestimation for layered dishes—and outline practical improvements (better fusion, depth cleaning, and targeted augmentations).

II. METHODOLOGY

This section describes in detail the selection and design of the model architecture, processing the original dataset, and the specific implementation of the training process.

A. Overall Architecture

The core design concept of the architecture originates from a fundamental principle in nutrition: the total calorie content of a food is determined by its physical volume and its energy density per unit volume. This physical relationship can be formalized as follows.

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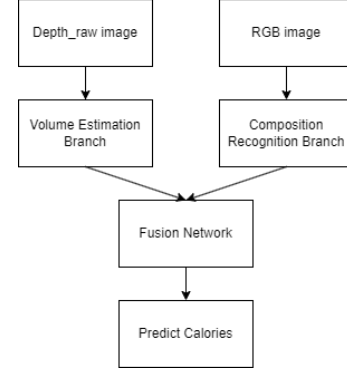


Fig. 1: Model flow diagram

$$\text{Calories} = \text{Volume} * \text{EnergyDensity} \quad (1)$$

Based on this principle, we propose a new architecture based on a two-stream neural network: the volume-composition network. The specific flow of the model is shown in Fig. 1. The model has three core modules: Volume branch, Composition branch and Fusion network. The two branches are trained separately and then fused to predict the value.

B. Volume Estimation Branch

The Volume branch takes a depth image as input, specifically designed to capture geometric features. The architecture includes:

Input: Depth image (1 channels, 224×224)

Convolutional block 1: 64 filters (5×5 kernel) + batch normalization + ReLU + 2×2 max pooling

Convolutional block 2: 128 filters (3×3 kernel) + batch normalization + ReLU

Convolutional block 3: 256 filters (3×3 kernel) + batch normalization + ReLU

Convolutional block 4: 512 filters (3×3 kernel) + batch normalization + ReLU + global average pooling

Output: 512-dimensional feature vector

Some design choices:

A larger convolution kernel in the first block provides a broader receptive field for capturing global geometric shape.

The 2×2 pooling layer efficiently reduces spatial resolution while retaining the essential volumetric structure.

The progressive channel expansion from 64 to 512 allows the network to learn increasingly abstract geometric representations relevant to food volume estimation.

C. Composition Recognition Branch

The composition branch processes RGB images to identify food types and ingredients. Its architecture is as follows.

Input: RGB image (3 channels, 224×224)

- Convolutional block 1: 64 filters (3×3 convolution kernel)
- + batch normalization + ReLU + 2×2 max pooling
- Convolutional block 2: 128 filters (3×3 convolution kernel)
- + batch normalization + ReLU + 2×2 max pooling
- Convolutional block 3: 256 filters (3×3 convolution kernel)
- + batch normalization + ReLU + 2×2 max pooling
- Convolutional block 4: 512 filters (3×3 convolution kernel)
- + batch normalization + ReLU + global average pooling
- Output: 512-dimensional feature vector

Some design principles:

The standard convolution kernel size (3×3) is suitable for capturing local texture and color patterns that characterize different types of food.

The 2×2 Pooling keeps more spatial information for detailed texture analysis and ingredient identification.

The symmetric channel matching the volume branch ensures balanced feature representation.

This design enables the network to distinguish foods with visual similarity but nutritional difference.

D. Fusion and Regression

To integrate geometric and compositional cues, a hybrid fusion strategy is employed that combines adaptive weighting and nonlinear interaction. A learnable parameter α controls the relative contribution of each modality. Given the 512-dimensional embeddings from both branches, they are first projected into a shared 256-dimensional space:

$$v' = \text{FC}_v(\text{Volume}), \quad c' = \text{FC}_c(\text{Composition})$$

The fusion vector is then constructed as:

$$\text{Fused} = [\alpha \cdot v', (1 - \alpha) \cdot c', v' \odot c']$$

where $\odot$ denotes element-wise multiplication. This formulation allows both explicit weighting and multiplicative cross-modal interaction.

The fused 768-dimensional feature is passed through a multilayer perceptron (MLP) for nonlinear regression:

- Fully Connected Layer 1: 768 → 512 with Batch Normalization, ReLU, and Dropout (0.4)
- Fully Connected Layer 2: 512 → 256 with ReLU
- Output Layer: 256 → 1 (calorie value)

Some design principles:

The learnable α parameter adaptively balances geometric and compositional contributions. The inclusion of the element-wise product $v' \odot c'$ explicitly models higher-order dependencies between modalities. Batch normalization and dropout enhance generalization and stabilize the regression of continuous calorie values.

E. Data Preprocessing and Augmentation

1. Input Preprocessing

Resizing: All images were resized to 224×224 pixels.

Normalization: RGB images were normalized using the ImageNet statistic with specific mean and standard deviation.

Depth Image Processing: Raw depth images were converted to a 1-channel format and similarly normalized as RGB images.

Splitting dataset: The original dataset was splitted into train set and validation set with a ratio of 8:2.

2. Data Augmentation

During training, we applied the following augmentations: random horizontal flips, random rotations, and color jittering on brightness, contrast, saturation and hue. No augmentation was performed during validation to ensure fair evaluation.

3. Data Cleaning

After checking the initial CSV file, we found some outliers, such as unusually large values and zeros. The extreme large values were removed. However, for the zeros, these images were all taken in sunlight, and we found similar images in the test set. Therefore, to maintain data integrity, these zeros were not removed.

F. Training Process

Number of training epochs: 60.

Validation frequency: Once per epoch.

Early Stop: If the performance does not improve in 10 epochs, stop training to avoid performance loss and overfitting.

Checkpoint: Save the best model based on validation loss.

Evaluation indicator: Use MSE as the primary indicator, also recording RMSE, MAE, and relative error.

Training completion: Plot the loss curves, RMSE curve, and MAE curve, and evaluate the best model.

III. EXPERIMENTATION AND EVALUATION

A. Baseline

Architecture: Two 4-layer CNN branches: a volume branch taking depth (geometric cues) with larger early kernels and aggressive pooling, and a composition branch taking RGB (appearance cues) with standard 3×3 blocks. Each branch outputs a 512-D vector (via AdaptiveAvgPool), which are concatenated and regressed by an MLP (1024→512→256→128→1). Early stopping with patience=5

Baseline performance:

TABLE I: Baseline performance

Setup	Volume Corr	Composition Corr	R^2	MSE
Baseline	-0.022	0.741	0.730	17135

B. Optimization Strategy

The optimization follows two strategies:

- Parameter tuning in the convolutional layer (volume)
- Strategy optimization in the fusion layer

C. Experiments

1. Wider channels (64 128 256 512 $\rightarrow$ 128 256 512 1024)

Purpose: To enhance the capability of volume estimator, a wider channel is applied to the convolutional layers in the volume estimator.

Action: All convolutional layers were doubled in width, while keeping the overall architecture (kernel sizes, pooling, and fusion) unchanged.

Performance:

TABLE II: Experiment 1

Setup	Volume Corr	Composition Corr	R^2	MSE
Wider channels	0.011	0.367	0.680	20762

Interpretation: Performance degraded, suggesting that the model overfit the limited training data. The excessive channel width likely led to optimization instability and sensitivity to pooling, suppressing the sensitive spatial cues.

2. Increased Patience (5 $\rightarrow$ 10)

Purpose: To prevent premature convergence by allowing the model more epochs to refine feature representations before early stopping.

Action: The early stopping patience was increased from 5 to 10 epochs, enabling the optimizer to continue training after temporary validation plateaus.

Performance:

TABLE III: Experiment 2

Setup	Volume Corr	Composition Corr	R^2	MSE
Increased Patience	0.115	0.677	0.802	12022

Interpretation: A clear improvement in generalization is observed (R^2 increased by 0.07 and MSE improved by 30%). Both volume and composition branches show stronger predictive correlations, indicating that the baseline model was not trained to its optimal performance before the early stop. The extended patience allows better convergence without severe overfitting, suggesting this modification enhances training stability.

3. Alpha Fusion

Purpose: To introduce an adaptive weighting mechanism between the volume and composition branches, enabling the network to learn the relative importance of geometric and visual cues automatically.

Action: A learnable parameter α was added to control the fusion balance between the two branches:

$$fused = \alpha \cdot v + (1 - \alpha) \cdot c$$

where v and c denote the respective feature embeddings.

Performance:

TABLE IV: Experiment 3

Setup	Volume Corr	Composition Corr	R^2	MSE
Alpha Fusion	0.473	0.720	0.797	10168

Interpretation: The model achieved a further 15% improvement in MSE, along with consistent gains across both correlation metrics. This indicates that the learned representations became more interpretable. The adaptive fusion mechanism allowed the network to place greater emphasis on the composition branch, which shows stronger correlation with caloric content. However, the slight drop in R^2 suggests that the linear fusion formulation may constrain the expressive interaction between modalities.

4. MLP Fusion Layer

Purpose: To enhance the interaction between the volume and composition features, a nonlinear fusion layer that explicitly models cross-modality relationships is introduced to the fusion layer.

Action: Extended the alpha fusion by concatenating $\alpha \cdot v$, $(1 - \alpha) \cdot c$, and their element-wise product $v \cdot c$, followed by a multilayer perceptron (MLP) for nonlinear combination:

$$fused = [\alpha \cdot v, (1 - \alpha) \cdot c, v \cdot c]$$

This design allows the model to learn higher-order dependencies between modalities.

Performance:

TABLE V: Experiment 4

Setup	Volume Corr	Composition Corr	R^2	MSE
MLP Fusion Layer	-0.204	0.738	0.808	9270

Interpretation: The nonlinear fusion slightly improved MSE by 10%, indicating stronger predictive performance. However, the negative volume correlation suggests that the fusion network over-relied on the composition branch while suppressing geometric cues. This imbalance implies that the MLP may have learned spurious cross-modality interactions, dominating by texture-based signals. Future work could regularize or reweight the fusion to better preserve volumetric information.

5. Fixed Alpha = 0.7

Purpose: To test the hypothesis that emphasizing the volume (depth) branch could improve calorie estimation by strengthening its correlation with food volume.

Action: The learnable α parameter was replaced with a fixed value of 0.7, manually biasing the fusion toward the volume features while keeping other network components unchanged.

Performance:

TABLE VI: Experiment 5

Setup	Volume Corr	Composition Corr	R^2	MSE
Fixed Alpha = 0.7	-0.233	0.717	0.711	12906

Interpretation: The model exhibited substantial performance degradation across all evaluation metrics. This decline can be attributed to the over-weighting of the volume branch, as the depth data in this dataset are prone to noise and illumination inconsistencies, leading to unstable volumetric representations. The observed negative correlation reinforces the limitation of depth-driven estimation and highlights the necessity of an adaptive, learnable fusion strategy to maintain modality balance.

IV. ERROR ANALYSIS

To further understand the behavior of our model, we conducted a detailed error analysis on the validation set. Figure 2 illustrates 10 representative examples with the smallest and largest prediction errors, displaying the RGB input, corresponding depth map, and predicted versus ground-truth calorie values.

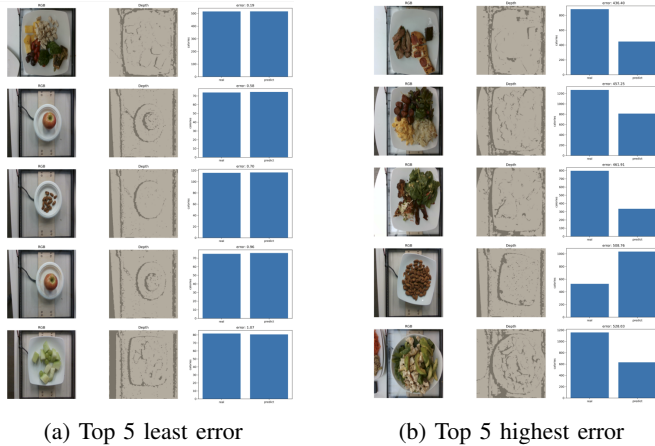


Fig. 2: Comparison of top 5 least and highest error predictions.

A. Successful Cases

The model performs well on small-portion and geometrically simple foods such as nuts, apples, and similar items in Fig. 2a. In these cases, the enhanced depth preprocessing produced smooth, consistent depth gradients that effectively capture overall volume. The composition branch also contributes strong cues when color and texture are discriminative, resulting in accurate calorie predictions. The adaptive α -fusion mechanism allowed the network to emphasize RGB or depth information depending on which modality was more reliable, contributing to stable generalization across diverse samples.

B. Failure Cases and Causes

As shown in the rows of Fig. 2b, the model tends to underestimate calories for complex, multi-layered dishes. These errors are primarily due to depth underestimation in occluded regions, where the sensor fails to capture the full geometry of overlapping food layers. From a theoretical standpoint, CNNs operating on pseudo-3D depth representations lack true volumetric reasoning and cannot infer self-occluded regions,

leading to systematic under-prediction of food volume and consequently total energy [1].

Illumination inconsistencies and specular reflections further impair the model’s ability to infer food volume by distorting the perceived geometry and shading cues in both RGB and depth inputs. When reflective surfaces or bright highlights occur, depth sensors often misinterpret reflected light as closer surfaces, resulting in underestimated depth values. Similarly, saturated RGB regions lose meaningful shading gradients, weakening the cues that guide volumetric reasoning in the fusion network. These effects collectively lead to systematic underestimation of food volume and thus caloric content.

C. Future Improvements

To mitigate these limitations, future work could integrate 3D shape reconstruction or depth completion networks to recover occluded geometry [2]. Incorporating cross-modal transformers or attention-based fusion could further enhance the interaction between RGB and depth modalities [3] [4]. Additionally, uncertainty-aware regression heads may allow the model to express prediction confidence and handle ambiguous or noisy inputs more robustly.

Overall, the error analysis highlights that while the proposed method captures general volumetric and compositional trends effectively, its accuracy degrades in geometrically challenging cases. Addressing these issues would likely yield significant gains in calorie estimation reliability.

V. CONCLUSION

This work presented a dual-branch convolutional framework for calorie estimation from paired RGB and depth images. The proposed method explicitly separates geometric and compositional information through dedicated branches and fuses them adaptively to predict caloric content. Systematic experiments demonstrated the effectiveness of the proposed design, reducing the mean squared error (MSE) on the test set to 9.3×10^3 , corresponding to substantial improvement over the baseline. Error analysis revealed that underestimation primarily occurs in multi-layered or occluded dishes due to incomplete depth capture. Future extensions such as depth completion, attention-based fusion, and uncertainty-aware regression are expected to further enhance robustness and generalization.

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